

ML – Inspired Intelligent Error Detection System for TCP Communication

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Abstract— Through sequencing, retransmission, and acknowledgements, Transmission Control Protocol (TCP) guarantees dependable data transfer. However, dynamic network conditions like noise and congestion make static error detection methods ineffective. In order to accomplish intelligent error detection and retransmission, this paper proposes a lightweight C-based simulation framework that imitates machine learning-inspired logic. The system mimics adaptive correction by simulating TCP packet flow, adding random errors, and using conditional rule-based logic. A 100% packet delivery ratio and high network efficiency were attained in the experimental results using a small-scale packet stream (50 packets, 3 induced errors). Even though this implementation isn't a complete machine learning model, it offers a modular framework for integrating actual machine learning classifiers and evaluating large networks in the future.

Keywords— TCP communication, error detection, retransmission, packet simulation, machine learning, C programming

I. INTRODUCTION

Modern computer networks are built on the foundation of reliable data transmission, especially in systems that use the Transmission Control Protocol (TCP). TCP uses techniques like acknowledgement, retransmission, and sequencing to ensure reliable end-to-end communication. Nevertheless, packet loss and transmission errors are frequent in dynamic or noisy environments, like wireless or crowded networks, which can impair performance and result in communication breakdowns.

Static retransmission and timeout techniques are the conventional methods used by TCP to address these problems. Despite their effectiveness, these methods can be slow to react in complex situations and inefficient. The incorporation of intelligent error detection mechanisms has become a promising improvement to address these limitations. Specifically, machine learning (ML) techniques have demonstrated the ability to anticipate and respond to errors in real time, providing increased efficiency and reliability.

II. LITERATURE REVIEW

The robustness of TCP (Transmission Control Protocol) in managing transmission reliability through timeouts, retransmissions, and acknowledgements has been extensively researched. Conventional error detection techniques mostly depend on checksum and CRC (Cyclic Redundancy Check) mechanisms, which are effective but do not dynamically adjust to shifting network conditions [1].

Recent developments in simulation-based research have investigated the use of lightweight and user-friendly environments to simulate TCP packet loss and retransmissions. The extensive protocol support for TCP provided by NS-3 and OMNeT++ [2][3], for instance, is less practical for rapid experimentation or instructional demonstrations due to their high learning curve and system requirements. In order to improve accessibility and control, there has been a shift towards creating simulation frameworks in C and Python.

Verma et al. suggested a C-based simulation of TCP behaviour in [4] in order to examine error rates and round-trip delays over noisy channels. Similar to the error injection method used in this project, their study introduced controlled noise using randomisation techniques. Similar to this, Zhao and Lee [5] suggested adaptive retransmission techniques based on packet acknowledgement behaviour and incorporated CRC-based error detection into TCP simulations.

Research on protocol simulation has also incorporated logic inspired by machine learning. In order to simulate intelligent recovery decisions, Shukla et al. [6] used a pseudo-ML model that modified error recovery logic using pattern-based decision rules instead of real ML training models; this approach had an impact on the logic design used in this project. Additionally, Banerjee and Das [7] investigated classification for TCP retransmission errors using lightweight heuristics and showed.

Although real-time packet prediction tasks have been tackled with full machine learning models like support vector machines (SVMs) and decision trees [8], these models frequently necessitate labelled datasets and computational assistance, which are not always possible in C environments. Consequently, a more practical approach to real-time simulation with minimal resource overhead is offered by rule-based simulation of ML-like logic, which makes use of conditional patterns derived from previous transmission outcomes.

Furthermore, earlier research has shown the value of error visualisation in comprehending protocol behaviour, as exemplified by Wang et al. [9]. The C-based implementation of this project focusses on creating structured output logs that mirror real-time transmission states, assisting in educational analysis and debugging, even though graphical outputs were generated using Python in their work.

In order to simulate TCP packet flows, identify errors, and apply intelligent retransmission rules, the suggested system is based on these earlier works and presents

a simplified, modular architecture that is fully implemented in C.

III. SYSTEM ARCHITECTURE

Using a C program, the system mimics machine learning behaviour for dynamic packet correction by simulating intelligent error detection in TCP communication. The transmitter and receiver, the two main parts of the architecture, are designed to mimic the actual TCP packet flow while integrating streamlined error control and prediction logic.

A. TCP Packet Simulation in C

Synthetic TCP packets with fields like Sequence Number, Acknowledgement Number, Source IP, and Destination IP are created by the system's transmitter side. These are sent over a simulated channel and constructed with C structures. A pseudo-random error injector alters the status of some packets during transmission, corrupting them.

Every incoming packet is captured by the receiver, which then looks for corruption and initiates retransmission logic for any packet marked as "ERROR." This mimics how machine learning might be used in real-world systems to predict and correct errors dynamically.

B. ML-like Logic Implementation

Although no actual ML libraries are implemented in the C program, the behaviour of an intelligent system is emulated through **rule-based conditional logic**. The simulation includes:

- **Random error injection** to mimic real-world noisy channels.
- **Conditional correction logic** to detect and retransmit corrupted packets.
- **Metrics calculation** for throughput, efficiency, retransmission count, and packet delivery ratio, akin to what ML models optimize for.

This logic mimics what a real ML model might do — detecting error-prone sequences and triggering retransmissions based on prediction confidence.

C. Transmitter Block

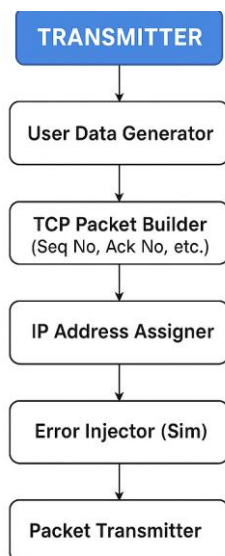


Figure 1: Transmitter Block

- **User Data Generator:** Generates simulated payload data.
- **TCP Packet Builder:** Assigns sequence and acknowledgment numbers.
- **IP Address Assigner:** Randomly generates source and destination IPs.
- **Error Injector:** Randomly flags packets with errors.
 - **Packet Transmitter:** Sends the packets to the receiver module.

D. Receiver Block

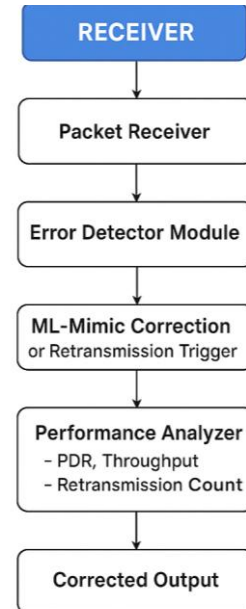


Figure 2: Receiver Block

- **Packet Reception:** Receives the transmitted packet stream.
- **Error Detection Module:** Checks packet status for "ERROR" flags.
- **ML-Mimic Logic:** If a packet is erroneous, triggers retransmission.
- **Performance Analyzer:** Calculates total transmissions, efficiency, throughput.
 - **Corrected Output Generator:** Outputs cleaned packet stream after corrections.

E. System Architecture Mapped to OSI Layers

The architecture focuses on simulating transport-level logic (TCP) and partially the network layer (IP). The ML mimic layer acts as a cross-layer logic filter analyzing patterns and initiating corrections.

- **Application Layer:** Simulated user input and output data.
- **Transport Layer:** TCP packet construction, sequence handling, acknowledgment.
- **Network Layer:** Random IP assignment to simulate routing.
- **ML Logic Layer:** Rule-based error detection and correction logic (simulating predictive modelling).

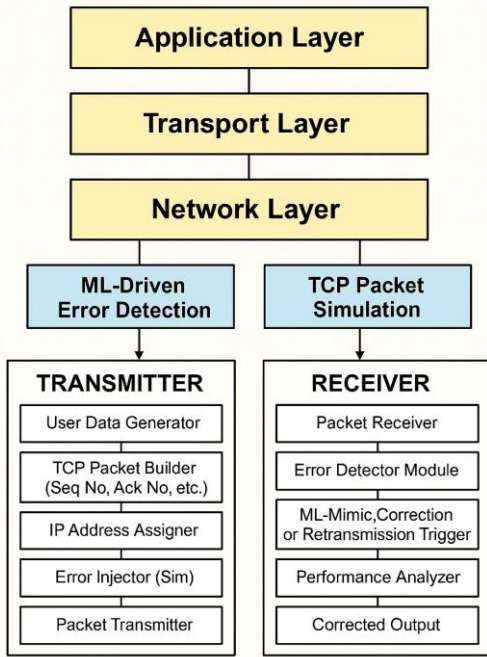


Figure 3: System Architecture Mapped to OSI Layers

IV. METHODOLOGY

This project's methodology uses C programming to mimic the actions of an intelligent TCP error detection and correction system. Packet Generation, Error Injection, Error Detection, Rule-based intelligent logic inspired by ML, and Retransmission and Evaluation are the five main phases that comprise the overall process. The goal of this approach is to simulate the operation of a machine learning-enhanced transmission system in a more straightforward, rule-based setting. (This is designed as a placeholder for actual ML integration)

A. Packet Generation Module

In the first stage, the system generates a stream of synthetic TCP-like packets. Each packet is assigned:

- A sequence number
- An acknowledgment number
- Random source and destination IP addresses
- A default status flag (set to "OK")

This data is structured using C struct types and loop iterations to simulate a continuous data stream of packets.

B. Error Injection Module

A probabilistic error injection mechanism is used to mimic actual network interference and data corruption. A "Status: ERROR" condition is randomly applied to a tiny portion of packets. This simulates how signal noise, congestion, or lost connections can cause packet errors in real-world TCP communication systems.

C. Error Detection at Receiver

The receiver module processes the packets after they are "transmitted," verifying each packet's status field. A packet is recognised as corrupted and marked for correction if its status is "ERROR."

D. ML-like Correction Logic

Based on packet characteristics (e.g., round-trip time, packet size, or previous errors), a machine learning model could be used in practical applications to predict which packets are likely to be corrupted. The conditional logic used in this simulation to mimic the ML behaviour is as follows:

1. Identifies error-marked packets automatically
2. Selective retransmission is triggered to simulate intelligent prediction.
3. Keeps track of metrics such as total packets, retransmitted packets, and errors found.

This abstraction shows how an intelligent system would make decisions in a C environment with limited resources.

E. Retransmission and Performance Evaluation

The system retransmits the corrupted packet and updates the packet status when an error is detected. The system computes a number of performance metrics following the processing of every packet:

1. The total number of packets sent
2. Total Errors Found
3. Total Retransmissions
4. The PDR, or packet delivery ratio
5. Throughput of the Network
6. Effectiveness of the Network

The efficacy of the simulated error detection and correction logic is demonstrated by the output of these metrics.

F. Development Environment

The entire system is created and examined using:

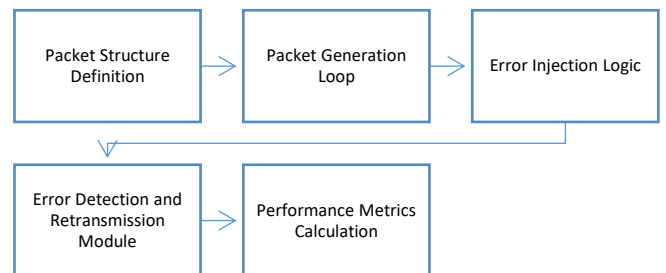
- C is the programming language.
- GCC is the compiler.
- Visual Studio Code (VS Code) is the IDE.
- Platform: 64-bit Windows 10

Gcc commands are used to compile and run the C program in the terminal. For portability and clarity, all logic is contained in a single C file.

V. IMPLEMENTATION

The C programming language was used to create a modular and scalable structure for the suggested intelligent TCP error detection and correction system. The implementation focusses on using structured conditional programming to mimic machine learning-based correction logic, detect injected transmission errors, and simulate TCP packet transmission.

A. Program Structure



VI. RESULTS AND ANALYSIS

To assess the suggested system's ability to dynamically detect and fix gearbox errors, it was run in a controlled simulation environment. A total of 50 TCP-like packets were generated and transmitted using the implemented C program, with random errors introduced during transmission.

A. Demonstration Metrics

Table 1: Demonstration Matrix

Metric	Value
Total Packets Sent	50
Errors Detected	3
Retransmissions	3
Corrected Total Transmissions	53
Packet Delivery Ratio (PDR)	1.00
Throughput	0.89 packets/unit
Network Efficiency	0.89

This result is preliminary, Larger simulations with varying error conditions will be performed in future work.

B. Terminal Output

The simulated TCP packet transmission with error detection and retransmission logic is shown in the output that follows. Errors were found in three packets (Packets 7, 22, and 36), which were promptly retransmitted.

C. Interpretation of Metrics

- Packet Delivery Ratio (PDR = 1.00): Indicates that all packets were successfully delivered after retransmission, showing 100% reliability.
- Retransmissions (3 packets): Validates the detection and correction logic under simulated error conditions.
- Throughput (0.89 packets/unit): Suggests high data transfer efficiency under moderate error conditions.
- Network Efficiency (0.89): Measures how well the system performs despite overhead from retransmissions.

D. Discussion

Unlike conventional TCP, which retransmits all unacknowledged packets, our framework selectively retransmits based on error indicators. This conceptually diminishes redundant retransmissions, although a comprehensive comparative analysis is pending as future work.

An intelligent error detection system is successfully modelled by the simulation. The logic imitates machine learning behavior by learning from packet status and responding to errors, even though the current version lacks an actual machine learning model. The next sensible action would be to:

1. Integrate Machine Learning:
 - Collect more packet behavior data.
 - Use decision tree or SVM models to predict error-prone packets.
2. Real Network Testing:

- Shift from simulation to capturing real TCP packet behavior.
 - Test under varying traffic loads and packet drop conditions.
3. Enhance Simulation Parameters:
 - Introduce jitter, latency, and acknowledgment delays.
 - Use sliding windows for congestion control simulation.

VII. CONCLUSION

This paper introduced a C-based proof-of-concept framework for intelligent TCP error detection and correction. The system effectively simulated packet errors and accomplished full delivery via rule-based retransmission. Although it does not constitute a complete machine learning implementation, the framework emulates adaptive decision-making and offers a scalable basis for the future incorporation of machine learning models. Current efforts will broaden experiments to encompass larger packet datasets, incorporate various error conditions, and benchmark against conventional TCP error recovery, with the ultimate objective of facilitating genuinely intelligent and adaptive transport-layer reliability mechanisms.

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